

Total No. of Questions : 8]

SEAT No. :

PC2671

[Total No. of Pages : 2

[6354]-836

B.E. (Electronics & Telecommunication)
INDUSTRIAL IN ROBOTICS & AUTOMATION
(2019 Pattern) (Semester - VII) (404181HR)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Draw neat diagrams wherever necessary.
- 3) Figures to the right indicates full marks.
- 4) All questions carry equal marks.
- 5) Assume suitable data, if necessary.

- Q1) a)** Explain basic components of pneumatic system with neat diagram. **[8]**
b) How pneumatic compressors are classified. Explain positive displacement compressors. **[9]**

OR

- Q2) a)** Explain air treatment stages and pressure Regulation method with neat diagram. **[8]**
b) Define pneumatic Actuators how they are classified. Explain Rotary actuators with neat diagram. **[9]**

- Q3) a)** State design features of CNC system. Explain drive system for CNC machine tools. **[9]**
b) Explain in detail how high speed automation insertion is possible in robotic assembly. State general rules for product design and automation. **[9]**

OR

- Q4) a)** Write steps to design of parts for high speed feeding and orienting. State different feeding difficulties. **[9]**
b) How to perform the analysis of an assembly. Explain with a suitable example. **[9]**

P.T.O.

- Q5) a)** Define Mechatronics. Explain basic elements of Mechatronics system with sketches. **[8]**
- b)** Prepare a detailed case studies on A Pick and place robot. Write pick and place robot algorithm. **[9]**

OR

- Q6) a)** What are the stages in mechatronics design. Explain different phases of mechatronics design process. **[8]**
- b)** Write a case study on The Engine Management system. Compare Traditional and mechatronics design. **[9]**
- Q7) a)** Compare Electrical, Mechanical and Hydraulic transmission systems. **[6]**
- b)** Draw neat diagram to achieve Forward and Reverse stroke using cylinder and piston mechanism. **[6]**
- c)** What is back pressure? When it arise and how to avoid it? **[6]**

OR

- Q8) a)** Draw and explain basic components of hydraulic system? **[6]**
- b)** Give comparison between hydraulic and pneumatic system. What are the applications of hydraulic system. **[6]**
- c)** Give classification of hydraulic Pumps. Explain positive displacement pumps and its characteristics. **[6]**

